Sahaj Bhadja

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Summary

Software engineer with 2+ years building scalable cloud applications, agentic AI workflows, secure APIs, and data-intensive services using Python, Java, React, TypeScript, SQL/NoSQL databases, automated testing, and performance-focused design.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Algorithms, Databases, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant AI services on Azure SQL and Blob Storage for 500+ users, optimizing schemas, access controls, and indexed queries to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant FastAPI orchestration for 1,000+ monthly LLM jobs using explicit states, retries, and failure isolation, tripling throughput while strengthening telemetry, debugging, and overall production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Python APIs and React workflows integrating GPT-4o, Claude, Qwen-VL, and OCR to handle 10,000+ daily requests while maintaining stable production performance and 90%+ evaluation consistency.
- Designed concurrent AWS RDS-backed processing for 1,000+ files per batch with secure ingestion and fault-tolerant execution, reducing turnaround by 60% while improving API reliability under peak load.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed cross-platform Python microservices and 20+ automated PyTest and Mockito suites through Jenkins CI/CD, improving release quality while supporting integrations across Windows, Linux, and macOS environments.
- Optimized REST API and ETL services using object-oriented patterns, AWS Glue, S3, and SQL, reducing response times by 30% and improving distributed data-pipeline efficiency by 68% overall.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN platform with REST APIs, WebSockets, and eight modular Node.js services, supporting 2M+ synchronized records while translating complex product requirements into maintainable real-time application workflows.
- Automated Docker-based testing and deployments through Jenkins CI/CD, reducing release time by 50% while improving code quality, documentation, and defect resolution across frequent customer-facing product updates.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Gathered client requirements and built Flask APIs for booking, scheduling, payment status, and user workflows across 50,000+ records, presenting system architecture and 12+ integration points before implementation.
- Integrated Firebase synchronization, JSON webhooks, and automated payment workflows, reducing manual processing by 70% while improving operational metrics, troubleshooting visibility, and stakeholder communication around recurring production issues.

Skills

Languages & Design: Python, Java, C++, JavaScript, TypeScript, SQL, Bash, object-oriented programming, data structures, algorithms, system design, concurrency, multithreading; **Applications & Data:** React.js, Node.js, FastAPI, Flask, REST APIs, WebSockets, microservices, distributed systems, SQL Server, PostgreSQL, MySQL, Azure SQL, MongoDB, data modeling, ETL, real-time processing; **AI & Cloud:** LLM orchestration, GPT-4o, Claude, Qwen-VL, LLaMA-2, RAG, embeddings, prompt engineering, evaluation, AWS, Azure, GCP, S3, Docker, Kubernetes; **Quality:** Jenkins, Git, CI/CD, PyTest, Mockito, unit testing, functional testing, test automation, debugging, telemetry, monitoring, performance optimization, fault tolerance, Agile/Scrum

Projects

Big-Brain Live | *Python, LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*
- Built an AI document assistant for 200+ users with secure uploads, semantic search, source-grounded summarization, and collaborative workflows, maintaining 95% query accuracy while reducing analysis time by 60%.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Node.js*
- Built an AI-powered database workspace supporting 10+ systems, combining schema-aware SQL generation, data exploration, and cross-platform deployment to accelerate enterprise application development across Windows, macOS, and Linux.

FinNexus Live | *Next.js, TypeScript, MongoDB, WebSockets, Kubernetes*
- Built a scalable platform for 50,000+ users and 1M+ transactions, combining secure APIs, real-time synchronization, advanced filtering, and Kubernetes-backed deployments for reliable high-volume enterprise customer workflows.